Q.1 What is 2 tailed and tailed T Test ?

Ans:- A t-test is a statistical method used to determine whether there is a significant difference between sample mean(s) and a population mean, especially when the sample size is small and population standard deviation is unknown.

One-Tailed t-Test

A one-tailed t-test is used when the alternative hypothesis specifies a **particular direction** of difference. It checks whether the sample mean is either greater than or less than the population mean.

Types of One-Tailed Tests:

1. Right-Tailed Test:

- H_0: $\mu = \mu_0$
- H_1: $\mu > \mu_0$
- Used when we want to check if the sample mean is significantly greater than the population mean.

2. Left-Tailed Test:

- H_0: $\mu = \mu_0$
- H_1: $\mu < \mu_0$
- Used when we want to check if the sample mean is significantly less than the population mean.

Characteristics:

- Rejection region lies on one side of the distribution.
- More powerful when direction is correctly specified.

Q.2 Explain what is Type I and Type II errors with an example.

Ans:- In hypothesis testing, decisions are based on probabilities, and there is always a possibility of making errors.

Type I Error (α Error)

A Type I error occurs when a true null hypothesis is rejected.

Explanation:

- We conclude that there is a significant effect when actually there is none.

Probability:

- Represented by α (level of significance)

Example:

Consider a medical test:

- H_0: Patient is healthy
- If the test incorrectly shows that the patient is diseased → Type I error

This is also known as a false positive.

Type II Error (β Error)

A Type II error occurs when a false null hypothesis is not rejected.

Explanation:

- We fail to detect a real effect or difference.

Probability:

- Represented by β

Example:

- H_0: Patient is healthy
- If the test incorrectly shows that the patient is healthy when actually diseased → Type II error

This is also known as a false negative.

Q.3What is power of statistical test?

Ans:- The power of a statistical test is defined as the probability of correctly rejecting a false null hypothesis.

Mathematical Expression:

Power = 1 – β

Where:

- β = probability of Type II error

Interpretation:

- High power means the test is effective in detecting true differences.
- Low power means there is a high chance of missing real effects.

Factors Affecting Power:

1. Sample Size:

- Larger sample size increases power.

2. Effect Size:

- Greater difference between means increases power.

3. Significance Level (α):

Higher α increases power but also increases Type I error.

4. Variability:

- Lower variability in data increases power.

Importance:

- Helps in designing experiments.
- Ensures reliability of statistical conclusions.
- Reduces chances of Type II errors.

Q.4 Explain input and out put characteristics for stats.ttest_rel function in your own words ***bold text***

Ans:- The function stats.ttest_rel() from the SciPy library is used to perform a paired sample t-test.

Purpose:

It compares two related datasets to determine whether their means differ significantly.

Input Characteristics:

1. Two Input Arrays:

- a: First dataset
- b: Second dataset

2. Conditions:

- Both datasets must have the same number of observations.
- Observations must be paired (e.g., before–after measurements).

3. Nature of Data:

Data should be continuous and approximately normally distributed.

Output Characteristics:

The function returns two values:

1. **t-statistic**

- Represents the magnitude of difference between paired samples.
- Larger absolute value indicates stronger evidence against H_0.

2. **p-value**

- Represents probability of obtaining observed result under null hypothesis.
- Used for decision making.

**Decision Rule:*

- If p-value ≤ α ⟶ Reject H_0
- If p-value > α ⟶ Accept H_0

Practical Interpretation:

This function helps determine whether changes between two related conditions (such as before and after treatment) are statistically significant or due to random variation.

Q.5 What is function in stats module for identifying power of a statistical test. Explain out put range of it and its significance. ***bold text***

Ans:-

Input Parameters:

- **effect_size:** Magnitude of difference between groups
- **nobs:** Number of observations (sample size)
- **alpha:** Level of significance

Output Range:

- The output value of power lies between 0 and 1

```
from statsmodels.stats.power import TTestPower

analysis = TTestPower()

power = analysis.power(effect_size=0.5, nobs=30, alpha=0.05)

print(power)
```

```
0.7539647157437905
```